

Need of S/W Engineering

- Large S/W
- Scalability
- Cost
- Dynamic nature
- Quality Management

S/W Components

- Program
 - Documentation
 - Operating Procedures
 - Code
 - Data
 - User interface
 - Libraries → Pre-written code
 - ~~Design~~ Test cases
 - Configuration files
 - Build & development scripts
 - Metadata
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S/W Characteristics

- functionality (Ability to perform according to specification)
- Usability (Ease of use)
- Efficiency (Utilization of time, effort, CPU, memory etc)
- Flexibility (Ability to adapt future changes)
- Reliability (Ability to work without fail over a period of time)
- Maintainability (How easily it can be repaired)
- Portability (Ability of S/W to work on different H/W Platforms without modification)
- Integrity (Secured from Hackers and cannot be modified with authorization)

Software Crisis

- Inability to develop desired S/W product
- Problems arisen during S/W development period.

Reasons :-

- Over budget
- over time
- ~~Unavailable~~ ~~slow~~ Unreliable S/W
- Difficulty in maintaining code
- Duplicate effort (Lack of automation)
- Lack of understanding

- Solutions :-
- Reduction in budget
 - Quality S/W development
 - Proper time-scheduling
 - Experienced developers
 - S/W must be deliverable

Significance of Automation

S/W Product

It is a S/W system delivered to customer with documentation that describes how to install and use the system.

Types

(1) Generic Products : - Stand-alone system that are developed by a producer unit and sold on the open market to any customer who is able to buy them. eg: - OS

(2) Customized Product : - System that are commissioned by a particular customer eg: - ERP

Characteristics

- (1) efficiency
- (2) maintainability
- (3) Dependability
- (4) on-time
- (5) Budgetary
- (6) Adaptability
- (7) functionality

Job responsibilities of programmer

(4)

Key responsibilities of a S/W engineer

- (1) Development
- (2) Analysis
- (3) Maintenance
- (4) Innovate
- (5) Planning and Project mgmt.

S/W engineer Roles & Responsibilities

(1) Junior S/W engineer

- maintaining sync betⁿ distributor and supplier
- Grade Program development on JScrip, HTML etc
- Develop a reporting system.
- Use agile methodologies
- undertake unit testing on project modules
- Develop user friendly design.

(2) Senior S/W engineer

- Develop S/W architecture based on requirements
- Create codes using effective tools & applications
- ad other resources and keep them confidential
- Recruit, train and mentor the project staff working under them
- stay updated with the prevalent technologies & trends.
- complete documentation of the developed S/W and run significant tests while adhering to the project deadlines.

Job

(3) Tech lead

- set the tech direction of the project
- identifying and adhere to the framework of the standards and procedures.
- compile reports and work in sync with other S/W engineers.
- facilitate communication in the organisation
- improve the development performance of the S/W using monitoring metrics.

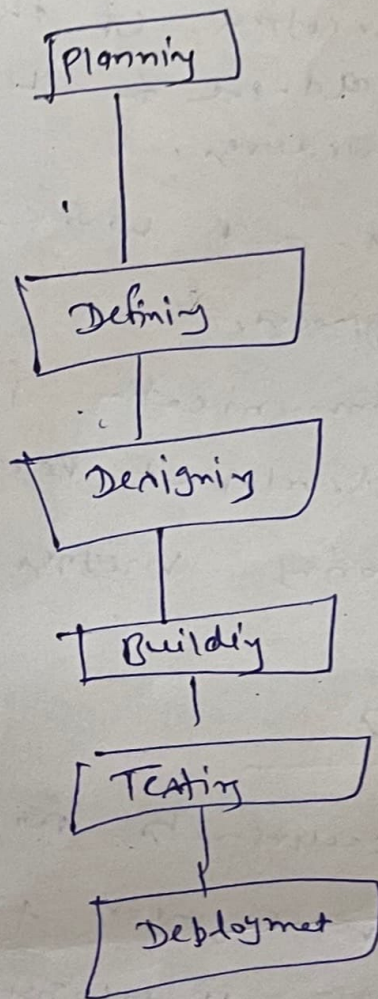
(4) S/W architect

- identifying the requirements for the new S/W / app
- work with other engineers to fabricate the software
- use tools and methodologies to create the desired user interface of the product / app
- Develop high level product specifications by focusing on the feasibility and compatibility of the product

Requirements and skills for a S/W Engineer

- front and backend coding knowledge
- understanding of agile development methodology
- Knowledge of testing and debugging
- writing clean code with clear comments

Software Development Life cycle



- ① Planning and Requirement Analysis
- ② Defining Requirements
- ③ Designing the Product Architecture
- ④ Building or developing the product
- ⑤ Testing the product
- ⑥ Deployment in the market and Maintenance